

Part 2 — Analysis of Five Documented RTL Bugs

Executive summary

This report examines five documented RTL failures and the verification gaps that allowed them to survive: (1) TL-UL response data/integrity mismatch, (2) lost OTP ECC error during multi-read assembly, (3) local FSM error priority blocking LC_CTRL escalation, (4) ROM_CTRL request logic mixing live and registered source selection, and (5) DMA chunk bookkeeping that skips bytes after a partial transfer.

The bugs span distinct classes. Two concern integrity/error observability; two concern control selection and priority; one concerns byte-accurate progress accounting. Their common theme is that local RTL expressions can violate transaction-level invariants across cycles, parallel data paths, or simultaneous events.

Bug	Public tracker status reviewed 9 Oct 2026	Local reduced test evidence
1 — TL-UL integrity	Open; no linked fix shown	Buggy model reproduces mismatch; fixed model passes
2 — OTP ECC capture	Closed; fix PR #5176	Source/issue-based; no local reproducer claimed
3 — LC_CTRL escalation	Closed; fix PR #12218	Issue/DV reproduction described; no local reproducer claimed
4 — ROM_CTRL select	Closed; fix PR #6523	Issue/DV reproduction described; no local reproducer claimed
5 — DMA chunk accounting	Open; no linked PR shown	Buggy model skips bytes 6–7; fixed model passes

The local tests are reduced behavioral models, not production-RTL simulations. Bugs 2–4 are supported by public tracker evidence but are not claimed as locally reproduced by the current test suite.

Method

For each issue, the report records the source, intended invariant, RTL mechanism, counterexample, likely escape path, a class-level assertion template, and detection costs/limitations. Assertion snippets are templates: adapt signal names, reset behavior, timing, and semantics to the actual implementation and specification.

Bug 1 — TL-UL response data and integrity become inconsistent

Source: OpenTitan issue #31187 —

<https://github.com/lowRISC/opentitan/issues/31187>

Component: `hw/ip/tlul/rtl/tlul\_fifo\_sync.sv`. Status: issue open in the reviewed source snapshot; no linked fix shown.

Intended behavior

Intended behavior: response payload and integrity metadata must describe the same value. If the payload changes, integrity must change consistently or the interface must explicitly exempt it from checking.

RTL mechanism

Mechanism: the response FIFO forwards `d\_data` only for `AccessAckData`; for other opcodes it substitutes zero. A separate integrity FIFO forwards `d\_user`, including `data\_intg`, unchanged. Thus `(data, ECC(data))` can leave as `(0, ECC(data))`. The issue notes current producers happen to use 0 or all-ones on non-data responses, for which this inverted-Hsiao code yields the same `7'h2A`; this is a convention, not a locally enforced contract.

Minimal counterexample

Counterexample: input `d\_opcode=AccessAck`, `d\_data=32'h42032080`, `data\_intg=7'h45`; output data becomes zero while integrity remains `7'h45`. The issue reports $ECC(0)=7'h2A$, so the pair is inconsistent.

Reduced SystemVerilog illustration

Reduced SV illustration:

```
always_comb begin
    d_data_o = (d_opcode_i == AccessAckData) ? d_data_i : 32'b0;
    data_intg_o = data_intg_i; // separate path, unchanged
end
// Required invariant: data_intg_o encodes d_data_o.
```

Independent related instance

Why it survives: tests may only use 0/all-ones for non-AccessAckData responses, hiding the mismatch. Scoreboards that mask don't-care response payloads can miss the transport-level integrity contract.

Detection property

Independent example: Ibex issue #2342 reports a distinct bug in which a store acknowledgement's don't-care payload can trigger a spurious store-response integrity alert. Same broad class—integrity logic applied without respecting transaction semantics—but not the same FIFO mux mechanism. <https://github.com/lowRISC/ibex/issues/2342>

Cost and limitations

Detection template:

Reproduction evidence / limitation

```
assert property (@(posedge clk_i) disable iff (!rst_ni)
  d_valid_o |-> data_integrity_ok_o);
```

Bug 2 — OTP correctable ECC error can be lost during multi-read assembly

Source: OpenTitan issue #5175 / fix PR

#5176: <https://github.com/lowRISC/opentitan/issues/5175>

Component: `otp\_ctrl\_dai.sv` and `prim\_generic\_otp.sv`. Status: issue closed and linked to PR #5176; this is a historical bug report, not a claim current upstream RTL remains faulty.

Intended behavior

Intended behavior: an ECC error from any sub-read contributing to a logical OTP word must be accumulated and reported when the logical operation completes.

RTL mechanism

Mechanism: the issue states that the DAI latches `otp\_err\_i` only when `otp\_rvalid\_i` is asserted. The generic OTP model assembles a 32-bit word from four narrower reads and asserts valid after the assembled read is ready. An error during one of the first three sub-reads can arrive before final valid and be missed.

Minimal counterexample

Counterexample: during a four-beat logical read, `otp\_err\_i` pulses on beat 1 while `otp\_rvalid\_i=0`. The valid-qualified latch does not remember it; final valid arrives later and status may omit the error.

Reduced SystemVerilog illustration

Reduced SV illustration:

Why it can survive verification

```
always_ff @(posedge clk_i or negedge rst_ni) begin
  if (!rst_ni) err_seen_q <= 1'b0;
  else if (otp_rvalid_i) err_seen_q <= otp_err_i;
end
// A pre-valid error is not accumulated by this logic.
```

Independent related instance

Why it survives: atomic 32-bit models and tests that inject ECC errors only on the final sub-read do not exercise the timing gap. The issue discussion identifies aggregation across block transfers as necessary.

Detection property

Independent example: OpenTitan EDN issue #19789 documents an error-response-policy gap in a separate IP: FIFO errors did not cause the same local escalation action as counter/FSM errors. Different mechanism, but a separate example of error detection and error response not being aligned.

<https://github.com/lowRISC/opentitan/issues/19789>

Cost and limitations

Detection template:

Reproduction evidence / limitation

```
assert property (@(posedge clk_i) disable iff (!rst_ni)
  (read_active_q && otp_err_i) |=> err_seen_q);
```

Bug 3 — LC_CTRL local FSM error blocks global escalation

Source: OpenTitan issue #12204 / fix PR #12218 <https://github.com/lowRISC/opentitan/issues/12204>

Component: OpenTitan life-cycle controller. Status: issue closed and links PR #12218; public issue records a failing DV scenario.

Intended behavior

Intended behavior: when global escalation is asserted, the controller must follow the specified escalation behavior even if a local FSM error occurs concurrently, unless the security specification defines another precedence.

RTL mechanism

Mechanism: the issue shows the FSM checking ``state_invalid_error`` or ``token_if_fsm_err_i`` first and selecting ``InvalidSt``. Only in the ``else if`` branch are ``esc_scrap_state0_i`` and ``esc_scrap_state1_i`` considered for ``EscalateSt``. Concurrent local error and escalation therefore select the earlier branch.

Minimal counterexample

Counterexample: ``esc_scrap_state0_i=1`` and ``token_if_fsm_err_i=1`` in the same decision cycle. The shown ordering selects ``InvalidSt``, so the global escalation is not reflected as ``EscalateSt`` in that decision. The issue reports a life-cycle state scoreboard mismatch.

Reduced SystemVerilog illustration

Reduced SV illustration:

Why it can survive verification

```
if (local_fsm_error) begin
    next_state = InvalidSt;
end else if (global_escalation) begin
    next_state = EscalateSt;
end
// Wrong if global escalation must dominate.
```

Independent related instance

Why it survives: single-fault tests pass: local error alone selects invalid, escalation alone selects escalation. Only simultaneous-fault injection exposes the precedence conflict.

Detection property

Independent example: OpenTitan ROM_CTRL issue #6516 is a separate IP's control-selection/transition bug, where live and registered source selection were mixed during reset/wakeup handover. It is not an identical FSM priority bug, but motivates explicit transition and concurrent-condition assertions.

<https://github.com/lowRISC/opentitan/issues/6516>

Cost and limitations

Detection template:

Reproduction evidence / limitation

```
assert property (@(posedge clk_i) disable iff (!rst_ni)
    global_escalation | => state_q == EscalateSt);
```

Bug 4 — ROM_CTRL request logic mixes live and registered selection

Source: OpenTitan issue #6516 / fix PR #6523 —

<https://github.com/lowRISC/opentitan/issues/6516>

Component: `hw/ip/rom\_ctrl/rtl/rom\_ctrl\_mux.sv`. Status: issue closed and linked to PR #6523; the issue says the proposed change made the smoke test pass.

Intended behavior

Intended behavior: ROM request, grant, and response bookkeeping must correspond to a coherent selected request source for the active transaction.

RTL mechanism

Mechanism: the reported expression was `assign rom\_req\_o = sel\_i ? 1'b1 : bus\_req\_i;`, using live `sel\_i` while other control logic uses registered `sel\_q`. During source handover, especially around reset release, request can remain asserted or switch sources at the wrong time.

Minimal counterexample

Counterexample: a power-manager smoke test wakes from low power and releases reset while checker and bus requests contend. Request remains asserted across a source transition and the SRAM adapter's `rvalidHighReqFifoEmpty` assertion fires. The proposed historical expression was `assign rom\_req\_o = sel\_q ? sel\_i : bus\_req\_i;`.

Reduced SystemVerilog illustration

Reduced SV illustration:

Why it can survive verification

// Faulty expression reported in the issue:

```
assign rom_req_o = sel_i ? 1'b1 : bus_req_i;
```

// Historical proposed correction:

```
assign rom_req_o = sel_q ? sel_i : bus_req_i;
```

Independent related instance

Why it survives: the failure depends on reset/wakeup timing and source handover. Steady-state tests with only one requester active do not create the condition.

Detection property

Independent example: LC_CTRL issue #12204 describes a different control-priority conflict when local FSM error and global escalation are simultaneous. The RTL mechanisms differ, but both demonstrate that transition behavior and simultaneous conditions require explicit checks.

<https://github.com/lowRISC/opentitan/issues/12204>

Cost and limitations

Detection template:

```
assert property (@(posedge clk_i) disable iff (!rst_ni)
  rom_req_o |-> (sel_q ? checker_req_active : bus_req_i));
```

Bug 5 — DMA chunk progress advances by nominal width after a partial transfer

Source: OpenTitan issue #31191 —

<https://github.com/lowRISC/opentitan/issues/31191>

Component: `hw/ip/dma/rtl/dma.sv`. Status: issue open in the reviewed snapshot; no linked branch or pull request shown.

Intended behavior

Intended behavior: counters and addresses advance by bytes actually transferred, including a short final transaction inside a chunk.

RTL mechanism

Mechanism: `remaining\_bytes` is computed correctly, but `transfer\_byte\_q` and `chunk\_byte\_q` advance by `transfer\_width\_q` unconditionally. Source/destination addresses likewise advance using nominal transfer width. If only two bytes remain in a six-byte chunk and nominal width is four, the transfer writes two bytes but bookkeeping advances four.

Minimal counterexample

Counterexample: `TOTAL\_DATA\_SIZE=12`, `CHUNK\_DATA\_SIZE=6`,
`TRANSFER\_WIDTH=4`. Buggy transfers occur at addresses 0 (4 bytes), 4 (2), 8 (4), and 12 (2); bytes 6 and 7 are skipped. Fixed reduced model uses addresses 0, 4, 6, and 10 and covers offsets 0–11.

Reduced SystemVerilog illustration

Reduced SV illustration:

Why it can survive verification

```
actual_bytes = (remaining_bytes < TRANSFER_WIDTH)
    ? remaining_bytes : TRANSFER_WIDTH;
transfer_byte_q <= transfer_byte_q + TRANSFER_WIDTH; // wrong
chunk_byte_q  <= chunk_byte_q  + TRANSFER_WIDTH; // wrong
// Correct bookkeeping advances by actual_bytes.
```

Independent related instance

Why it survives: the issue explicitly says the DV sequence item constrained chunk sizes to multiples of transfer width. That excluded the failing configuration from random testing. This is a verification-space restriction, not proof that the RTL supports arbitrary chunk sizes.

Detection property

Independent example: LiteX release notes list a separate DMA empty/misaligned-transfer fix under commit identifier `59fb89991`. This is an independent DMA implementation and related boundary/alignment class, not evidence of the identical counter-update mechanism. <https://github.com/enjoy-digital/litex/releases>

Cost and limitations

Detection template:

Reproduction evidence / limitation

```
assert property (@(posedge clk_i) disable iff (!rst_ni)
    transfer_accept | => transfer_byte_q ==
    $past(transfer_byte_q) + $past(actual_bytes));
```

Cross-bug synthesis

FAILURE CLASS	EXAMPLE	INVARIANT
Integrity/data-path consistency	Bug 1	Payload and integrity metadata describe the same value.
Error capture across multi-beat operation	Bug 2	An error on any sub-operation remains observable until completion.
Priority among simultaneous conditions	Bug 3	Security escalation precedence is explicit under concurrent faults.
Temporal consistency of request selection	Bug 4	Security escalation precedence is explicit under concurrent faults.
Byte-accurate progress accounting	Bug 5	Progress advances by actual accepted bytes, not nominal width.

Verification techniques

- Transaction-level assertions: relate payload to integrity, request to grant/response, and accepted bytes to progress counters.
- Temporal assertions: make error aggregation and escalation explicit across cycles; include simultaneous conditions and reset/wakeup transitions.
- Constrained-random coverage: do not constrain away legal boundary cases unless the RTL/spec explicitly prohibits them.
- Formal verification: prove invariants over legal interleavings and size configurations; justify assumptions from the interface specification.
- Mutation/regression tests: show buggy-model failure and fixed-model pass, and keep reproducers small enough that the failure mechanism is obvious.

Costs and limitations

Most properties are low-cost assertions, but value depends on correct temporal semantics. Sampling only final valid can miss an earlier error pulse; assuming one requester can hide a source handover; assuming aligned DMA chunks proves only a restricted configuration space. Formal does not replace a precise specification. Overly broad assumptions can make proofs vacuous; overly strict assertions can flag legal don't-care fields.

What the reduced models establish

For Bug 1, the local reduced model reproduced a data/integrity mismatch and the fixed model passed. For Bug 5, the local reduced model reproduced the missing-byte pattern for 12/6/4 and the fixed model transferred offsets 0–11. Bugs 2–4 are supported by public issue reports and historical fix links, but are not claimed as locally reproduced. If the assignment requires every bug to be independently reproduced, runnable minimal reproducers and captured logs for Bugs 2–4 remain outstanding.

Search log and source list

Search date: 20september-8october 2026. Process: searched public project trackers using issue numbers and distinctive signal names; inspected issue bodies for source locations, counterexamples, fix links, and tracker status; searched for independent examples from other IP/projects; then mapped each mechanism to an assertion template.

- [1] OpenTitan TL-UL issue #31187, response data/integrity mismatch:
<https://github.com/lowRISC/opentitan/issues/31187>
- [2] OpenTitan OTP issue #5175 / PR #5176, correctable ECC error not reflected in status: <https://github.com/lowRISC/opentitan/issues/5175>
- [3] OpenTitan LC_CTRL issue #12204 / PR #12218, FSM error blocks escalation: <https://github.com/lowRISC/opentitan/issues/12204>
- [4] OpenTitan ROM_CTRL issue #6516 / PR #6523, bus grant/request mismatch: <https://github.com/lowRISC/opentitan/issues/6516>
- [5] OpenTitan DMA issue #31191, chunk size not multiple of transfer width: <https://github.com/lowRISC/opentitan/issues/31191>
- [6] Ibex issue #2342, possible spurious store-response integrity alerts: <https://github.com/lowRISC/ibex/issues/2342>
- [7] OpenTitan EDN issue #19789, FIFO error does not cause local escalation: <https://github.com/lowRISC/opentitan/issues/19789>
- [8] LiteX release notes, DMA empty/misaligned-transfer fix identifier `59fb89991`: <https://github.com/enjoy-digital/litex/releases>
- [9] OpenTitan TL-UL documentation: <https://opentitan.org/book/hw/ip/tlul/index.html>
- [10] OpenTitan DMA theory of operation: https://opentitan.org/book/hw/ip/dma/doc/theory_of_operation.html

Appendix A — Search log and candidate disposition

Recordkeeping note. The working notes captured issue selection and the local Bug 1/Bug 5 runs, but did not preserve a timestamped query-by-query diary at the moment each search was made. The entries below are therefore a transparent reconstruction from the saved project notes and the follow-up source checks performed on 9 October 2026; they are not represented as a contemporaneous browsing transcript. For the next iteration, append each query and decision immediately when searching.

Search sequence and decisions

1. Screen the assignment's exclusion list before selecting candidates

Where: the exclusion list supplied in the assignment prompt, then the corresponding public GitHub issue pages for any candidate being considered.

Search/result: compare candidate topics against listed categories and exact excluded references (deadlock/livelock/starvation, CDC/metastability, reset/init, decode/spec conformance, pipeline control, arithmetic/width/parameterization, FIFO/buffering, FSM/access control, peripheral/analogue, and language-semantics examples).

Decision: do not reuse a listed case or merely rename the same excluded bug. In particular, OpenTitan #23822 (the listed FIFO `tf\_push` data-cycle issue) was set aside as explicitly excluded. A FIFO topic alone was not treated as excluded; the exact mechanism and issue identity were checked.

2. Earlier SYSRST candidate — OpenTitan #11565

Where: OpenTitan GitHub issue #11565, <https://github.com/lowRISC/opentitan/issues/11565>.

Search/result: the issue describes `intr\_status` only triggering once for a pin edge change because `q\_sync\_q` was not cleared after the first interrupt.

Decision: set aside as the primary Bug 1 choice in favour of #31187. The SYSRST issue is a real RTL bug, but the TL-UL candidate provides a more explicit data-path invariant, concrete payload/ECC values, identifiable RTL lines, and a compact reduced-model counterexample for this report. This is a suitability decision, not a claim that #11565 is invalid.

3. TL-UL data-zeroing candidates — OpenTitan #14611, #17330, and #31187

Where: OpenTitan issue tracker; specific issue pages:

<https://github.com/lowRISC/opentitan/issues/14611>,
<https://github.com/lowRISC/opentitan/issues/17330>, and
<https://github.com/lowRISC/opentitan/issues/31187>.

Search/result: #17330 discusses TL-UL data-zeroing inconsistencies; #31187 pinpoints `tlul\_fifo\_sync` zeroing non-`AccessAckData` response data while forwarding original integrity metadata, and gives the exact mux and an ECC counterexample.

Decision: use #31187 as the primary case because the issue directly identifies the offending expression and the integrity mismatch. Treat the other data-zeroing discussions as context/related history, not as additional independent bugs, to avoid double-counting a related issue family.

4. OTP error capture — OpenTitan #5175 / PR #5176

Where: <https://github.com/lowRISC/opentitan/issues/5175> and the linked fix PR #5176.

Search/result: the issue states ``otp_err_i`` is only latched when ``otp_rvalid_i`` is asserted, while the generic OTP wrapper assembles a word from four narrower reads; an earlier sub-read error may be missed.

Decision: retain as a historical bug because the timing mechanism is concrete and the issue links a fix. Mark it as source-based only in this report; no local reproducer was run in the current working session.

5. LC_CTRL escalation priority — OpenTitan #12204 / PR #12218

Where: [Where: https://github.com/lowRISC/opentitan/issues/12204](https://github.com/lowRISC/opentitan/issues/12204) and linked PR #12218.

Search/result: the issue documents ``token_if_fsm_err_i`` preventing the FSM from entering ``EscalateSt`` when scrap/escalation inputs are asserted.

Decision: retain because the faulty priority ordering is explicit and the report includes a plausible concurrent-fault assertion. The property remains a template pending signal/latency verification against the exact RTL; no local reproducer is claimed.

6. ROM_CTRL request selection — OpenTitan #6516 / PR #6523

Where: <https://github.com/lowRISC/opentitan/issues/6516> and linked PR #6523.

Search/result: the issue reports ``rom_req_o = sel_i ? 1'b1 : bus_req_i``, while the associated selection bookkeeping uses ``sel_q``; the failure appears during reset/wakeup source handover.

Decision: retain because the issue gives a specific expression, trigger sequence, and historical correction. It is distinct from the LC_CTRL priority case, although both require temporal/transition coverage.

7. DMA chunk accounting — OpenTitan #31191

Where: <https://github.com/lowRISC/opentitan/issues/31191> and the OpenTitan DMA theory-of-operation documentation.

Search/result: for total size 12, chunk size 6, and transfer width 4, the issue shows the second transaction of the first chunk writes two bytes but advances counters by four. The issue also explains that DV constrained chunk sizes to multiples of transfer width, excluding the failing case.

Decision: retain because it supplies exact configuration values, source lines, the missed-byte consequence, and a concrete explanation for why verification did not find it. A local reduced model reproduces the skipped offsets 6–7; it does not compile or simulate production ``dma.sv``.

8. Independent example for Bug 1 — Ibex #2342

Where: <https://github.com/lowRISC/ibex/issues/2342>.

Search/result: a store acknowledgement's don't-care payload can be passed through a read-data ECC decoder and raise a spurious store-response alert.

Decision: use only as a related independent example, not as the same mechanism. It shares the broader lesson that integrity logic must respect transaction semantics, but it is not the same FIFO payload mux.

9. Independent DMA example — LiteX release notes

Where: <https://github.com/enjoy-digital/litex/releases>.

Search/result: release notes mention a separate DMA empty/misaligned-transfer fix associated with identifier `59fb89991`.

Decision: use cautiously as a related boundary/alignment example only. It is not asserted to share OpenTitan's exact counter-update mechanism; the evidence is less detailed than the primary OpenTitan issue.

10. Final candidate check and local reproduction

Where: public issue pages listed in the source list; local macOS Terminal using Icarus Verilog 13.0 in the project working tree.

Search/result: inspected the exact Bug 1 and Bug 5 reduced-model files and ran their testbenches. Bug 1's buggy model reported a data/integrity mismatch and its fixed model passed. Bug 5's buggy trace used addresses 0, 4, 8, 12 and marked bytes 6–7 missing; the fixed trace used 0, 4, 6, 10 and transferred bytes 0–11.

Decision: Decision: claim local reduced-model reproduction only for Bugs 1 and 5. Do not claim that the production RTL was compiled or that Bugs 2–4 were independently reproduced.

Candidates and approaches set aside

- OpenTitan #23822: excluded because the assignment's supplied exclusion list names this FIFO `tf\_push` data-cycle issue. It was not revived as a new candidate.
- OpenTitan #11565 (SYSRST): real issue, but set aside in favour of the more directly checkable TL-UL integrity mismatch in #31187.
- OpenTitan #14611 and #17330: related TL-UL data-zeroing history; not counted as separate primary bugs alongside #31187 to avoid overlap/double-counting.
- Ibex #2342: not selected as a primary bug because the current task's chosen primary list already includes an integrity-path example; retained as a different-design comparison for Bug 1.
- LiteX DMA issue/fix: not selected as the primary DMA bug because the OpenTitan #31191 report provides more explicit source-level accounting, exact parameters, and an explanation of the DV constraint.
- Candidates in the assignment's expressly excluded deadlock, CDC, reset/init, decode, arithmetic/parameterization, and language-semantics categories were not selected merely to fill the five slots. The screening decision was based on the assignment's exclusions, not a claim that those categories are unimportant.

How to maintain this log going forward

For every new candidate, record the query, repository/page, access date, relevant source lines or commit, why it might qualify, the exact exclusion/overlap check, and the keep/reject decision before moving to the next candidate. Preserve rejected candidates even when the reason is simply weaker evidence, duplicate mechanism, or no runnable minimal counterexample.